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**DATABASE MANAGEMENT SYSTEM**  
DBMS ASSIGNMENT

**ONLINE SHOPPING MANAGEMENT SYSTEM**

*A Relational Database Design for Managing Online Shopping Operations*

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|----------------------|--------------------------------------------------|
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| <b>Course</b>        | BCA – Artificial Intelligence & Machine Learning |
| <b>Subject</b>       | Database Management System                       |
| <b>University</b>    | Alliance University                              |
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## OBJECTIVE

To design a complete Entity Relationship (ER) Diagram for the Online Shopping Management System and convert it into a Relational Schema by identifying entities, attributes, primary keys, foreign keys, relationships, and cardinalities.

## ENTITIES AND ATTRIBUTES

### Entity 1: CUSTOMER

| Column        | Key         |
|---------------|-------------|
| customer_id   | Primary Key |
| customer_name | -           |
| email         | -           |
| phone         | -           |
| address       | -           |

### Entity 2: CATEGORY

| Column        | Key         |
|---------------|-------------|
| category_id   | Primary Key |
| category_name | -           |
| description   | -           |

### Entity 3: PRODUCT

| Column         | Key         |
|----------------|-------------|
| product_id     | Primary Key |
| product_name   | -           |
| price          | -           |
| stock_quantity | -           |
| category_id    | Foreign Key |

|             |             |
|-------------|-------------|
| supplier_id | Foreign Key |
|-------------|-------------|

#### Entity 4: SUPPLIER

| Column        | Key         |
|---------------|-------------|
| supplier_id   | Primary Key |
| supplier_name | -           |
| email         | -           |
| phone         | -           |

#### Entity 5: ORDERS

| Column       | Key         |
|--------------|-------------|
| order_id     | Primary Key |
| customer_id  | Foreign Key |
| order_date   | -           |
| total_amount | -           |

#### Entity 6: ORDER\_ITEM

| Column        | Key         |
|---------------|-------------|
| order_item_id | Primary Key |
| order_id      | Foreign Key |
| product_id    | Foreign Key |
| quantity      | -           |
| unit_price    | -           |

### Entity 7: PAYMENT

| Column         | Key         |
|----------------|-------------|
| payment_id     | Primary Key |
| order_id       | Foreign Key |
| payment_date   | -           |
| payment_method | -           |
| payment_status | -           |

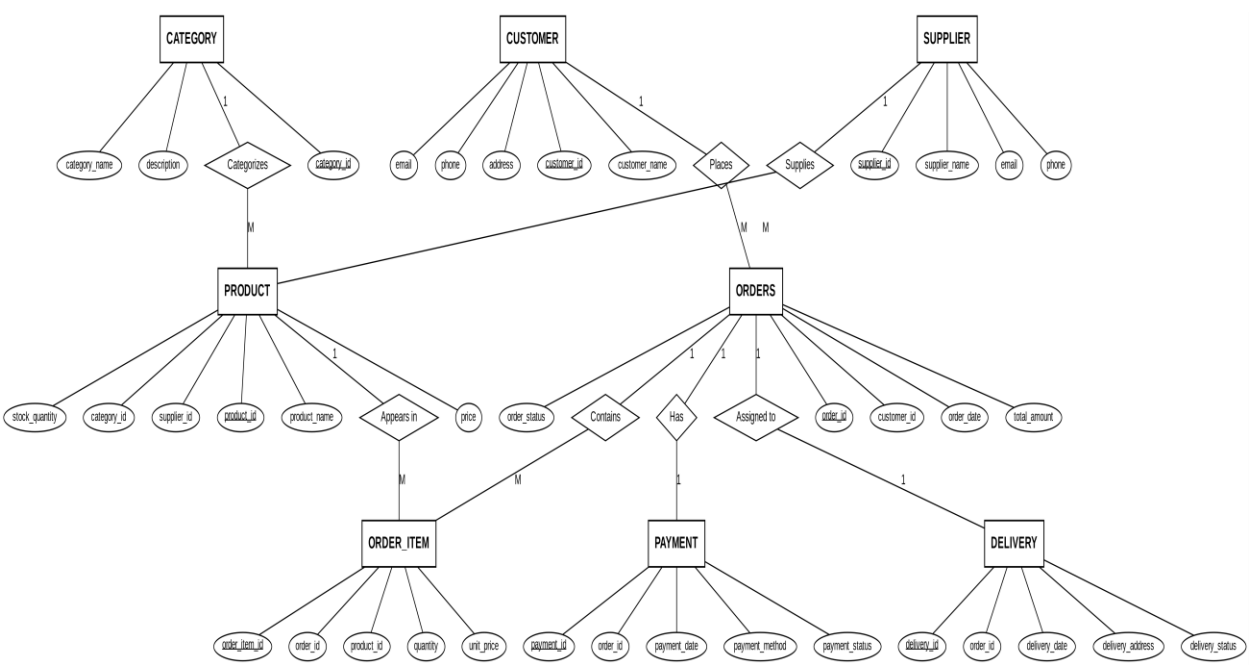
### Entity 8: DELIVERY

| Column           | Key         |
|------------------|-------------|
| delivery_id      | Primary Key |
| order_id         | Foreign Key |
| delivery_date    | -           |
| delivery_address | -           |
| delivery_status  | -           |

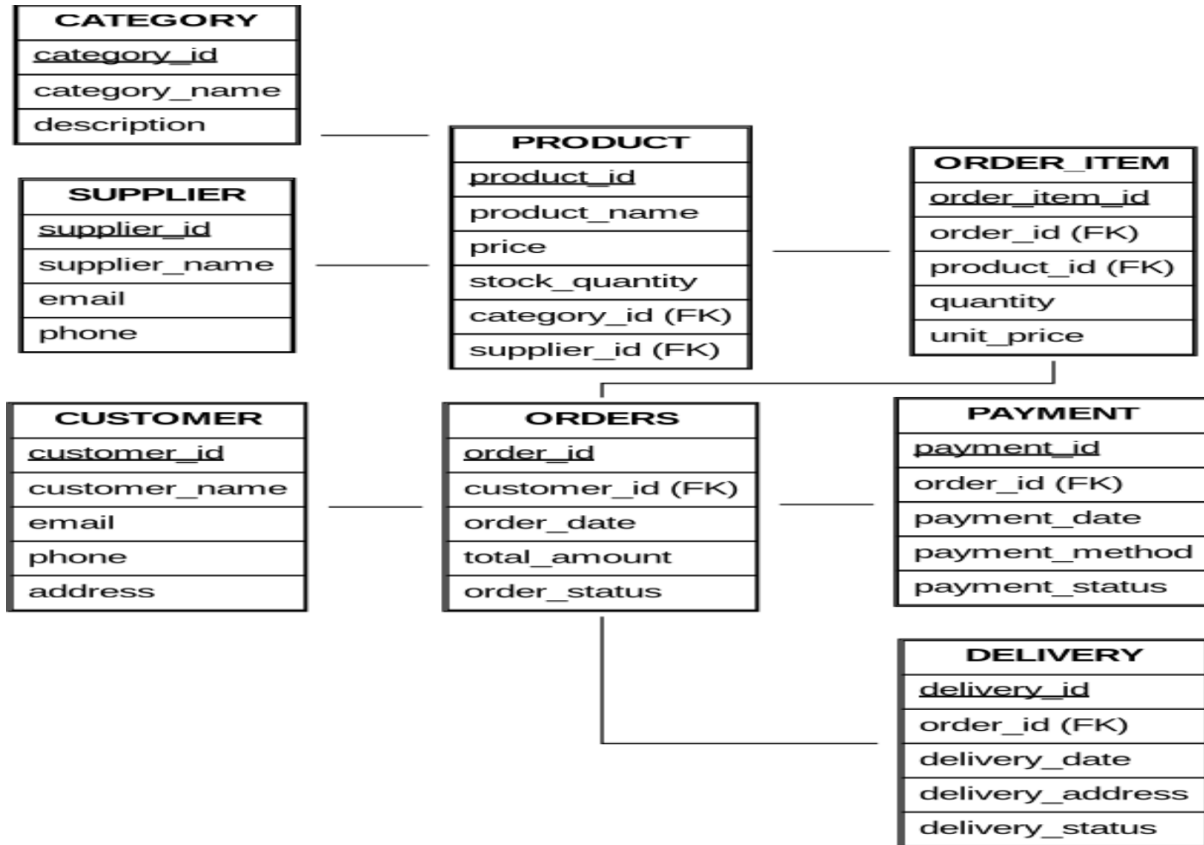
## RELATIONSHIPS AND CARDINALITIES

| Entity 1 | Relationship | Entity 2   | Cardinality |
|----------|--------------|------------|-------------|
| CUSTOMER | Places       | ORDERS     | 1:M         |
| CATEGORY | Categorizes  | PRODUCT    | 1:M         |
| SUPPLIER | Supplies     | PRODUCT    | 1:M         |
| ORDERS   | Contains     | ORDER_ITEM | 1:M         |
| PRODUCT  | Appears in   | ORDER_ITEM | 1:M         |
| ORDERS   | Has          | PAYMENT    | 1:1         |
| ORDERS   | Assigned to  | DELIVERY   | 1:1         |

# ER DIAGRAM



## RELATIONAL SCHEMA



## PRIMARY KEYS AND FOREIGN KEYS

| Table      | Primary Key   | Foreign Key              |
|------------|---------------|--------------------------|
| CUSTOMER   | customer_id   | -                        |
| CATEGORY   | category_id   | -                        |
| PRODUCT    | product_id    | category_id, supplier_id |
| SUPPLIER   | supplier_id   | -                        |
| ORDERS     | order_id      | customer_id              |
| ORDER_ITEM | order_item_id | order_id, product_id     |
| PAYMENT    | payment_id    | order_id                 |

|          |             |          |
|----------|-------------|----------|
| DELIVERY | delivery_id | order_id |
|----------|-------------|----------|

## RELATIONSHIP EXPLANATION

**Customer - Orders:** One customer can place many orders; each order belongs to one customer. Therefore, the relationship is 1:M.

**Category - Product:** One category can contain multiple products. Therefore, the relationship is 1:M.

**Supplier - Product:** One supplier can supply multiple products. Therefore, the relationship is 1:M.

**Orders - Order\_Item:** One order can contain multiple order-item records. Therefore, the relationship is 1:M.

**Product - Order\_Item:** One product can appear in many order-item records. Therefore, the relationship is 1:M.

**Orders - Payment:** Each order has one payment record in this simplified model. Therefore, the relationship is 1:1.

**Orders - Delivery:** Each order has one delivery record in this simplified model. Therefore, the relationship is 1:1.